

Trịnh Ngọc Các

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PERSONAL INTRODUCTION

I am a fresh Data Science graduate at the University of Economics Ho Chi Minh City, with an interest in researching the application and integration of AI for practical purposes, mainly in the fields of Computer Vision and Signal Processing. Due to limited practical experience in the development and deployment of AI systems, I am seeking opportunities to gain hands-on experience in these areas while further honing my existing knowledge.

I mainly use: Git, Python, Lua, Bash

EDUCATION

University of Economics Ho Chi Minh City
College of Technology and Design
Bachelor of Science in Data Science

Aug, 2022 – Mar, 2025

- Graduation Thesis: "A Framework for Radiology Image De-identification using Generative AI"
- Overall GPA: 3.86/4.0

WORK EXPERIENCE

AAI Labs
Researcher in Engineering Department, Remote

Vilnius, Lithuania
Aug, 2025 – Feb, 2026

- **Scope:** Responsible for the prototyping and development of AI-integrated solutions.
- **Details:**
 - I mainly take part in the brainstorming and development of a first version for said solutions. At AAI Labs, we use **Scrum** to manage and plan our projects.
 - Participated in the deployment of an traffic sign detection model for a dash-cam system, using YOLO-based models.
 - Investigated and developed segmentation algorithms to generate discount trends based on historical sales data.
- **Technologies Used:** Git, Flask, Flask-Restful, Pytest, Numpy, Pandas, OpenPyXL, Scikit-learn, PyTorch

PROJECTS

Music Genre Classification

Oct, 2025 – Present

- **Overview:** This is my current personal project to learn more about Signal Processing. Music Genre Classification is an elementary audio classification task in MIR. The project mainly delves into feature engineering methods using Short-time Fourier Transforms to obtain spectral information from the waveforms for downstream models to perform classification.
- **Methodology:** Currently, the project is developing common models such as deep CNN and ResNet. These models all classify samples using three main types of features: Chromagrams, pitch-based Log-Frequency Spectrograms and Mel-scale-based spectrograms. The current challenge remains to address the overfitting of models due to the limited data and the insufficient preprocessing strategy being used.
- **Libraries Used:** PyTorch
- **Link to repository:** <https://github.com/Ngoc-Cac/genre-classification>

Big Data Application for Sentiment Analysis on KFC Reviews in HCMC

Apr, 2025 – May, 2025

- **Overview:** This group project investigates the development and deployment of a Big Data system to support the storage and processing of KFC reviews in Ho Chi Minh City.
- **Methodology:** In this project, we used Big Data technologies such as Apache Hadoop and Apache Spark to store and process large amount of unstructured data. We followed a basic ETL process where reviews are extracted from Google Maps by means of web crawling into our HDFS deployed using pre-built Docker images. Afterwards, the data is preprocessed and saved for future analysis and model training with PySpark and PyTorch.
- **Responsibilities:** I am responsible for the main implementation of the application, including: building environment and service containers for analysis, reviews crawling functionality and model training functionality.
- **Technologies Used:** Docker, Apache Hadoop, Apache Spark, Selenium, PySpark, PyTorch, Transformers
- **Link to repository:** <https://github.com/Ngoc-Cac/kfc-reviews-with-bigdata>

Radiology Image De-identification using Generative AI

Aug, 2025 – Oct, 2025

- **Overview:** This project investigates the topic of removing sensitive information present within radiology images, using image segmentation and image inpainting. The project provides a proof of concept for a convenient and scalable de-identification pipeline, which can serve medical experts with no prior experience to generative AI.

- **Methodology:** The project combines Segment Anything Model with inpainting models such as LaMa and Stable Diffusion to perform the de-identification process. To investigate Stable Diffusion's performance on capturing anatomical information, I also fine-tuned the model on X-ray images provided by the MURA dataset.
- **Libraries Used:** PyTorch, Diffusers, Accelerate, bitsandbytes, Gradio
- **Link to repository:** https://github.com/Ngoc-Cac/gen_ai_deidentification

High-Performance Computing Weather Prediction System

Apr, 2025 – May, 2025

- **Overview:** This group final project experimented with the use of Docker Clusters to deploy a high-performance computing system that serves a weather prediction service.
- **Methodology:** We experimented with a cluster-based computing system using Docker Clusters to deploy two main services: a backend API serving our trained LSTM models and a frontend web UI for demonstration purposes. While the objective is to deploy a multi-node cluster, we decided to simulate this environment using VMs built with VirtualBox, due to limited budget.
- **Responsibilities:** My role in this project revolves around the backend functionalities, which includes the building and maintenance of necessary containers and clusters in order to deploy and run the application, the implementation and deployment of API endpoints needed in the frontend UI. I am also responsible for the training of the models used for weather prediction.
- **Technologies Used:** Docker, Docker Clusters, FastAPI, Chakra UI, PyTorch, Transformers
- **Link to repository:** <https://github.com/Ngoc-Cac/weather-prediction>

RESEARCH ACTIVITIES

Proceedings of STAIS national conference 2024

May, 2024 – Aug, 2024

Citation: Cac, T. N., Khanh D. T. M., Nguyen, T. N. T., Yen, N. V., Thanh D. N. H. (2024). *Rerouting Airline Passengers of Canceled Flights with Genetic Algorithms*. STAIS 2024

- **Role:** Lead author and presenter of paper.
- **Description:** The paper investigates the use of Genetic Algorithms to reroute passengers so that most passengers can still finish their journeys. We achieved this by modelling the problem as a Maximum Flow Problem, and applied a self-implemented genetic algorithm to compute the edge weights. We found the implemented algorithm capable of solving networks with roughly 20-50 edges in reasonable time.
- **Code for paper:** <https://github.com/Ngoc-Cac/MaxFlowGA>

AWARDS

UEH Young Researcher Awards 2025

Dec, 2024 – Apr, 2025

- **Dec, 2024 – Jan, 2025:** Preparation and submission of work titled "*Rerouting Airline Passengers of Canceled Flights with Genetic Algorithms*".
- **Jan, 2025:** Results announcement of Rank A.

LANGUAGE SKILLS

Vietnamese: Native

English:

- TOEIC 2-Skill: 965 overall (taken in 2024)
- IELTS: 8.0 overall (taken in 2021)
- Fluent in English, can converse on a daily basis.
- Intermediate-level academic writing.